Space Emanation Theory

Source, Capacity and the Luminosity History

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Rough draft 1 | 23 October 2026

Abstract

The SET source law treats a specified collection of matter as one source, whether that collection is a planet, a galaxy or a cosmological region. This paper develops the cosmological case from its density form. The source law supplies an intensive source-volume clock. When that clock is identified with the reference time in the capacity relation, the retained homogeneous source model reduces to one explicit equation for the growth of its source radius.

$$\frac{dR_h}{dT_Q} = \frac{2R_h}{3T_Q} + c\sqrt{1 - \frac{2R_h^2}{3c^2T_Q^2}} \tag{1}$$

The equation separates two contributions: the evolution of the material distribution and the advance of the source-accumulation front. Its exact reduction gives the burden history already used in SET's luminosity calculation. An independent integration of the source radius and photon distance reproduces all fifteen archived Pantheon+ and DES-Dovekie brightness contrasts, with a maximum numerical difference below 7.1×10^{-12} and no changed inputs or fitted coefficients.

The derivation also exposes the physical feedback. New source shells increase collective burden, while density evolution dilutes it. On the branch used for the luminosity calculation, the response approaches $S^2/c^2 = 6/7$. The paper follows this result through photon energy, arrival intervals and received brightness, and identifies the complete-energy relation needed to connect this cosmological realization to the common field prescription.

The main results

1. A source-volume clock independent of the chosen homogeneous subdivision.
2. An explicit source-radius equation and an exact integral for its history.
3. A direct reconstruction of the existing luminosity predictions.
4. A calculable accumulation-dilution balance and a precise complete-energy target.

1. One source law, one capacity relation

SET begins with the participation of source burden, translational motion and local event progression in one field capacity. For a clock moving relative to the local field reading,

$$S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2, \qquad d\tau = \beta\, dT \tag{2}$$

For the comoving clock used in this homogeneous calculation, the translational term vanishes. We write its progression as α and the central collective burden as S_c:

$$\alpha^2 = 1 - u, \qquad u \equiv \frac{S_c^2}{c^2}, \qquad dt = \alpha\, dT \tag{3}$$

The source law for a specified domain Ω is

$$Q_\Omega^2 = 24\pi G M_\Omega V_\Omega, \qquad Q_\Omega = M_\Omega \sqrt{\frac{24\pi G}{\rho_\Omega}} \tag{4}$$

Here ρ_Ω = M_Ω/V_Ω. For a sphere, V = 4πR³/3, so this is the same law written as Q² = 32π²GMR³. Q has units of volume per time. Changing the description from a planet to a disjoint collection of sources does not change the algebraic law. The specified domain and the location at which its response is read must remain explicit.

Two domains with different bookkeeping

The subscript D denotes a material collection with fixed baryonic mass. Its volume can evolve. The subscript h denotes the advancing spherical source domain; it acquires additional matter as its boundary moves through the distribution. The same law applies to both.

$$\frac{Q_D^2}{V_D} = 24\pi G M_D, \qquad 2\frac{dQ}{Q} = \frac{dM}{M} + \frac{dV}{V} \tag{5}$$

An unchanged source law therefore does not require an unchanged numerical Q when its domain volume or enclosed mass changes. The fixed-mass source strength is Q_D²/V_D. This distinction is necessary when continuous sourcing is followed through a growing cosmological domain.

2. Why the central reading contains 3GM/R

Take a uniform spherical source domain of physical radius R_h and density ρ at the time of the reading. The retained isotropic shell response gives a central burden contribution 2G dm/r from a shell at radius r. Thus

$$dm = 4\pi\rho r^2\, dr, \qquad S_c^2 = 2G\int_0^{R_h} \frac{dm}{r} \tag{6}$$

$$S_c^2 = 8\pi G\rho \int_0^{R_h} r\, dr = 4\pi G\rho R_h^2 = \frac{3GM_h}{R_h} \tag{7}$$

The coefficient 3 comes from including the shells throughout the source domain and reading their combined response at its centre. It is not a coefficient adjusted to supernova brightness. This step uses the isotropic source-response kernel retained in the homogeneous application, in addition to the finite-domain source law.

$$M_h = \tfrac{4\pi}{3}\rho R_h^3, \qquad S_b^2 = \frac{2GM_h}{R_h}, \qquad S_c^2 = \tfrac{3}{2}S_b^2 \tag{8}$$

The surface reading S_b and central reading S_c concern the same source but different locations. For this sphere, Q_h/(4πR_h²) gives S_b. Substituting the central S_c in that same surface formula would mix two different readings.

The growing-radius capacity scale

If density is held fixed while the source radius is enlarged, the central response rises linearly with radius:

$$S_c(R) = R\sqrt{4\pi G\rho}, \qquad R_{\text{cap}} = \frac{c}{\sqrt{4\pi G\rho}} \tag{9}$$

$$Q(R_{\text{cap}}) = \sqrt{\frac{2c^6}{3G^2\rho^2}} \tag{10}$$

This is the algebraic capacity scale suggested by the collective-source picture. An evolving history must also evolve density. The next steps determine what that distinction does to the result. A local equality S_c = c does not by itself establish which photons reach an observer; that requires the photon trajectory.

3. The clock supplied by the source law

For the conserved material collection D, introduce the source-volume variable T_Q by integrating the volume increment divided by its source rate. The positive expanding branch gives

$$dT_Q \equiv \frac{dV_D}{Q_D} = \frac{dV_D}{\sqrt{24\pi G M_D V_D}} \tag{11}$$

$$T_Q = \sqrt{\frac{V_D}{6\pi G M_D}} = \frac{1}{\sqrt{6\pi G\rho}}, \qquad \frac{dV_D}{dT_Q} = Q_D \tag{12}$$

The displayed origin is the formal zero-volume origin of this source-volume variable. Its differential is unchanged by a constant shift. At a given density, multiplying the representative domain's mass and volume by the same factor leaves T_Q unchanged. The construction therefore does not depend on the arbitrary size of a homogeneous subdivision.

$$\rho = \frac{1}{6\pi G T_Q^2}, \qquad H_b \equiv \frac{Q_D}{3V_D} = \frac{2}{3T_Q} \tag{13}$$

Since V_D is proportional to a^3 for a conserved homogeneous collection, its scale factor obeys

$$\frac{a}{a_0} = \left(\frac{T_Q}{T_{Q0}}\right)^{2/3} \tag{14}$$

When the enclosed mass changes

The advancing domain h contains new matter. Differentiating the same density-defined variable while keeping this change gives

$$dT_Q = \frac{dV_h - (V_h/M_h)\,dM_h}{Q_h} \tag{15}$$

The second term removes volume growth caused solely by enclosing additional mass at the same density. This is how the source clock distinguishes the evolution of existing material from the advance of the source boundary. At a zero-radius launch, T_Q is evaluated from the finite representative-domain density, rather than from the front's zero mass and zero volume.

4. Connecting the source clock to capacity

The source-volume identity is exact, but an operational variable must be connected to the physical clock used for motion. Let t denote the comoving material clock and H = (1/a) da/dt. For any monotone homogeneous volume history,

$$\alpha_Q \equiv \frac{dt}{dT_Q} = \frac{H_b}{H} \tag{16}$$

The SET cosmological realization used for the luminosity calculation identifies this source-volume clock with the reference clock in the capacity relation, and uses the collective central burden for that relation:

$$\frac{dt}{dT_Q} = \alpha_Q = \alpha = \sqrt{1-u}, \qquad u = \frac{4\pi G\rho R_h^2}{c^2} \tag{17}$$

This is the physical correspondence that closes the source-clock history. The finite-domain source law constructs T_Q; the capacity interpretation assigns the material clock's progression relative to it. Stating that correspondence at this step keeps the origin of the calculation visible.

$$\frac{dV_D}{dt} = \frac{Q_D}{\alpha}, \qquad H = \frac{H_b}{\alpha} = \frac{2}{3T_Q\sqrt{1-u}} \tag{18}$$

What Q normalized means here

The same sourcing process has a larger numerical rate when expressed per second of the slower material clock:

$$Q_{\mathrm{norm},\,D} = \frac{Q_D}{\alpha}, \qquad Q_D\, dT_Q = Q_{\mathrm{norm},\,D}\, dt \tag{19}$$

This equality introduces no extra source. It describes one increment using two clocks. The receiving photon need not carry a clock for the surrounding field and material observers to define these intervals. Its propagation is calculated with the same clock relation.

In the realization examined here, the physical material volume evolves with t, so a(t) is dynamical. The luminosity result below belongs to that specified history. The equations will determine how its burden and optical response combine.

5. A causal source front and its mass

Retain the homogeneous physical spatial reading $d\ell^2 = a^2 d\chi^2$ and the common limiting speed c relative to material clocks. An outward radial null front obeys

$$\frac{d\chi_h}{dt} = \frac{c}{a}, \qquad R_h = a\chi_h \tag{20}$$

Differentiating its physical radius gives the front equation used in the luminosity application:

$$\dot{R}_h = HR_h + c \tag{21}$$

A dot denotes d/dt. The term c is local front propagation; HR_h accounts for the changing physical spatial scale. This calculation identifies that null accumulation domain, launched from a specified initial event, with the source domain whose central burden sets local capacity. The central kernel is an equal-time uniform readout of that domain; a fully retarded inhomogeneous kernel is a separate field calculation.

Material conservation and boundary sweep

$$\dot{\rho} = -3H\rho, \qquad M_h = \frac{4\pi}{3}\rho R_h^3, \qquad V_h = \frac{4\pi}{3}R_h^3 \tag{22}$$

$$\dot{M}_h = 4\pi\rho R_h^2 c, \qquad \dot{V}_h = 3HV_h + 4\pi R_h^2 c \tag{23}$$

The first equality follows by differentiating M_h: dilution of the existing material cancels its contribution from the HR_h term, leaving the new mass swept up at the front. With $A_h = 4\pi R_h^2$, the source-volume form is

$$\frac{dV_h}{dT_Q} = Q_h + \alpha c A_h \tag{24}$$

The full moving volume therefore has both a material-evolution term and a swept-volume term. Applying $dV_h/dT_Q = Q_h$ to this entire advancing region would omit the latter. The law $dV_D/dT_Q = Q_D$ applies directly to the fixed material collection.

6. Accumulation, dilution and the radius equation

Now differentiate the central burden u = 4πGρR_h²/c². Material conservation and null-front growth give

$$\dot{u} = -Hu + \frac{8\pi G\rho R_h}{c} \tag{25}$$

$$\frac{\dot{u}}{u} = \frac{2c}{R_h} - H \qquad (R_h > 0) \tag{26}$$

This is the balance in the collective-source picture. The front includes new shells and raises the burden. Evolution of the material distribution dilutes it. The undivided equation remains regular at the zero-radius launch.

The source clock makes density explicit. Substitution in the central response gives

$$u = \frac{2R_h^2}{3c^2T_Q^2} \tag{27}$$

Multiplying the null-front equation by dt/dT_Q = √(1−u) and using H = 2/[3T_Q√(1−u)] now yields

$$\frac{dR_h}{dT_Q} = \frac{2R_h}{3T_Q} + c\sqrt{1 - \frac{2R_h^2}{3c^2T_Q^2}} \tag{28}$$

The source law, the central burden reading, the retained clock correspondence and the null front have become one evolution equation. It determines the front’s history once its initial state is supplied. No luminosity coefficient has been introduced in reaching it.

A dimensionless form

$$X \equiv \frac{R_h}{cT_Q}, \qquad \frac{dX}{d\ln T_Q} = -\frac{X}{3} + \sqrt{1 - \frac{2X^2}{3}} \tag{29}$$

This form is convenient for integration and for seeing the response’s limiting behavior. Its physical capacity range is 0 ≤ X < √(3/2).

7. The exact luminosity-history equation

The burden variable is u = 2X²/3, while N ≡ ln a satisfies dN = (2/3)d ln T_Q. Therefore

$$\frac{du}{dN} = 2X\frac{dX}{d\ln T_Q} = -u + \sqrt{6u(1-u)} \tag{30}$$

This is precisely the burden history used in the archived luminosity calculation. It has now been reached by following the source radius and the density-defined source clock. For a regular integration variable on the lower branch, write

$$y \equiv \sqrt{\frac{u}{1-u}}, \qquad \frac{dy}{dN} = \frac{(1+y^2)(\sqrt{6}-y)}{2} \tag{31}$$

For k = √6, separation uses the elementary partial fractions

$$\frac{2}{(1+y^2)(k-y)} = \frac{2}{7}\left[\frac{1}{k-y} + \frac{y+k}{1+y^2}\right] \tag{32}$$

Integrate from the zero-response front y_i = 0 at a_i. The exact result for 0 ≤ y < √6 is

$$\ln\frac{a}{a_i} = \frac{2}{7}\left[-\ln\frac{\sqrt{6}-y}{\sqrt{6}} + \frac{1}{2}\ln(1+y^2) + \sqrt{6}\arctan y\right] \tag{33}$$

Differentiating this expression recovers the separated equation. Once the present burden is fixed by the source inputs, it determines a_i/a_0 for the zero-front branch. The launch ratio is not an additional fitted parameter.

Preserving the original front at launch

The u equation alone is not Lipschitz at u = 0 and admits mathematical waiting branches. The original radius equation fixes immediate front growth: at R_h = 0 and finite T_Q, dR_h/dT_Q = c. The regular y equation preserves this choice. The radius or y equation should therefore be used to initiate the calculation.

8. The source quantities along the history

Once u(T_Q) is known, mass, radius and Q follow algebraically. Start with the central response and the uniform density:

$$\rho = \frac{1}{6\pi G T_Q^2}, \qquad R_h = cT_Q\sqrt{\frac{3u}{2}} \tag{34}$$

$$M_h = \frac{4\pi}{3}\rho R_h^3 = \frac{2R_h^3}{9GT_Q^2} = \frac{c^3 T_Q}{\sqrt{6}G}u^{3/2} \tag{35}$$

$$Q_h = 12\pi G T_Q M_h, \qquad Q_{\mathrm{norm},\,h} = \frac{Q_h}{\sqrt{1-u}} \tag{36}$$

These expressions show how the same source law follows the changing collective domain. Q_h grows as new matter enters and its defining volume changes. No new source species appears in these identities.

The frozen reception benchmark

Quantity	Value
M_h0	1.5005279083 × 10⁵³ kg
R_h0	4.3992396698 × 10²⁶ m = 46.5 billion light-years
ρ_0	4.2074852219 × 10⁻²⁸ kg/m³
Q_h0	5.1893886387 × 10⁶² m³/s
u_0; α_0	0.7598922704; 0.4900078873
T_Q0	43.55432725 Gyr in source-reference units
H_b0; H_0	14.96663; 30.54365 km s⁻¹ Mpc⁻¹

These are the previously frozen, cosmology-conditioned source benchmark inputs and their derived quantities. They are reused here unchanged; they are not a new independent measurement of the absolute source scale. The grouped brightness comparison below cancels its common distance normalization.

Independent radius integration recovers the reception radius and mass to relative differences of 1.7×10^{-14} and 5.0×10^{-14}. This checks the consistency of the source-radius reconstruction.

9. What the evolving collective source approaches

The dimensionless radius equation has one positive equilibrium. Solving its right-hand side equal to zero gives

$$X_* = \frac{3}{\sqrt{7}}, \qquad u_* = \frac{6}{7}, \qquad \alpha_* = \frac{1}{\sqrt{7}} \tag{37}$$

Below that equilibrium the burden grows; above it the burden decreases. The luminosity branch approaches S_c/c = √(6/7) ≈ 0.925820. Its clock remains real and positive. Fixed-density enlargement and evolution with changing density therefore give different limits under the same source law.

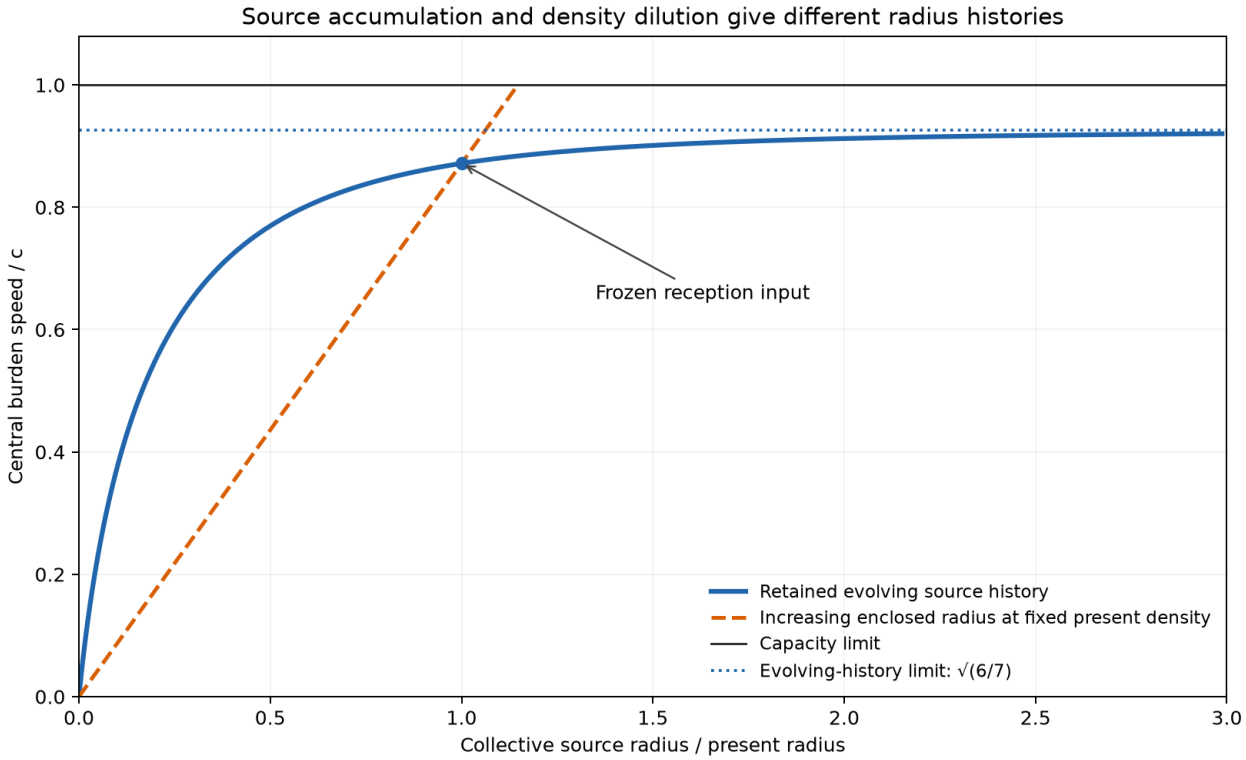


Figure 1. Model histories with the same reception benchmark. The fixed-density construction reaches the central capacity scale. The evolving source includes density dilution and approaches S_c/c = √(6/7). These curves are calculations, not observational data.

$$HR_h \longrightarrow 2c, \qquad \dot{R}_h \longrightarrow 3c, \qquad \rho \propto R_h^{-2} \tag{38}$$

The areal-radius rate 3c combines front propagation with the changing physical scale. The local propagation term remains c. At this limit M_h/R_h is constant. The evolution therefore regulates the central burden through the distribution’s density history, rather than requiring it to rise without limit.

10. From the clock history to photon brightness

Use the retained homogeneous photon coupling with comoving momentum p_χ. Its reference-time Hamiltonian and material-clock photon energy are

$$\mathcal{H}_\gamma = \alpha \frac{c|p_\chi|}{a}, \qquad E_\gamma = \frac{\mathcal{H}_\gamma}{\alpha} = \frac{c|p_\chi|}{a} \tag{39}$$

Spatial homogeneity conserves p_χ. Thus the photon energy ratio and the propagation interval give

$$\frac{E_o}{E_e} = \frac{a_e}{a_o} = \frac{1}{1+z}, \qquad \chi = c\int_{t_e}^{t_o} \frac{dt}{a} = c\int \frac{\alpha\, dT_Q}{a} \tag{40}$$

$$\frac{dt_o}{dt_e} = \frac{a_o}{a_e} = 1 + z \tag{41}$$

The arrival-interval relation follows by varying the endpoints of the same null path between comoving emitter and receiver. For conserved photons spreading over the flat homogeneous beam area 4π(a_oχ)², one factor of 1+z reduces each photon's energy and a second reduces the arrival rate:

$$F_o = \frac{L_e}{4\pi (a_o\chi)^2 (1+z)^2}, \qquad D_L = (1+z) a_o \chi \tag{42}$$

$$D_L(z) = (1+z) c \int_0^z \frac{dz'}{H(z')}, \qquad H(z) = H_{b0} \frac{(1+z)^{3/2}}{\alpha(z)} \tag{43}$$

The burden history enters through the clock-weighted distance integral. The source exposure identity Q dT_Q = Q_normalized dt is already respected in this calculation. An additional Q/α brightness multiplier after integrating this history would add a response that the photon derivation above has not supplied.

The physical content is specific: a collective source history changes clock progression, the history determines photon travel and arrival, and those quantities determine luminosity distance. Received brightness is obtained from that chain.

11. The observable and a worked calculation

The archived test compares standardized brightness between supernova groups at different redshifts. It does not compare the fading of one supernova at two dates. Each target group is referenced to the catalogue's nearby sample, using the same contrast weights and published uncertainties as the earlier calculation.

Define d_L = H_0D_L/c, with H_0 = H_b0/α_0. Let $\mathscr{L}$ be the target-minus-reference weighted contrast, with $\mathscr{L}[1] = 0$. The constant-clock history supplies the dimensionless baseline. With the catalogue's heliocentric redshift in the prefactor and Hubble-diagram redshift in the integral,

$$d_{L,0} = 2(1 + z_{\mathrm{HEL}})\left[1 - (1 + z_{\mathrm{HD}})^{-1/2}\right] \tag{44}$$

$$d_L = (1 + z_{\mathrm{HEL}})\int_0^{z_{\mathrm{HD}}} \frac{\alpha(z')}{\alpha_0(1+z')^{3/2}}\, dz' \tag{45}$$

$$B_{\mathrm{obs}} = 10^{-0.4\mathcal{L}[m - 5\log_{10} d_{L,0}]}, \qquad B_{\mathrm{SET}} = 10^{-0.4\mathcal{L}[5\log_{10}(d_L/d_{L,0})]} \tag{46}$$

A common additive magnitude offset and a common distance normalization cancel because the contrast has zero total weight. Predictions are evaluated at each catalogue redshift before the contrast is formed; a group's mean redshift is only its plotting label.

Pantheon+: mean redshift 0.730852

For the archived group of 63 target objects and its 311-object nearby reference, the baseline is B_0 = 1. The radius-derived history gives a contrast magnitude of 0.3769637728. The measured contrast is 0.3862377659. Therefore

$$B_{\mathrm{SET}} = 10^{-0.4(0.3769637728)} = 0.7066664790 \tag{47}$$

$$B_{\mathrm{obs}} = 10^{-0.4(0.3862377659)} = 0.7006560778 \tag{48}$$

The observed one-sigma interval is 0.679033 to 0.722968. The central discrepancy falls from 42.72% for the unit baseline to 0.86% for this SET calculation. This worked example uses the same rule as every group in the following table.

12. The fifteen reproduced brightness contrasts

The independent radius calculation reproduces the complete archived set below. $B_0 = 1$ for every group. The error is $100(B_{SET}/B_{obs} - 1)$; quoted intervals are the archived marginal one-sigma observational intervals.

Catalogue	Mean z	Observed [1σ]	SET	Error
Pantheon+	0.158	0.9198 [0.9024, 0.9375]	0.9370	+1.87%
Pantheon+	0.273	0.8428 [0.8272, 0.8586]	0.8790	+4.30%
Pantheon+	0.413	0.7798 [0.7667, 0.7931]	0.8170	+4.78%
Pantheon+	0.567	0.7444 [0.7268, 0.7623]	0.7587	+1.93%
Pantheon+	0.731	0.7007 [0.6790, 0.7230]	0.7067	+0.86%
Pantheon+	0.902	0.6720 [0.6194, 0.7290]	0.6613	-1.58%
Pantheon+	1.122	0.4941 [0.4546, 0.5371]	0.6141	+24.27%
Pantheon+	1.530	0.5639 [0.5237, 0.6072]	0.5512	-2.25%
DES-Dovekie	0.161	0.9547 [0.9307, 0.9794]	0.9358	-1.98%
DES-Dovekie	0.286	0.8572 [0.8405, 0.8743]	0.8732	+1.87%
DES-Dovekie	0.422	0.7899 [0.7711, 0.8092]	0.8138	+3.02%
DES-Dovekie	0.573	0.7332 [0.7193, 0.7474]	0.7574	+3.30%
DES-Dovekie	0.706	0.6940 [0.6808, 0.7075]	0.7146	+2.96%
DES-Dovekie	0.872	0.6624 [0.6446, 0.6808]	0.6691	+1.02%
DES-Dovekie	1.064	0.6567 [0.6280, 0.6866]	0.6254	-4.76%

All fifteen central discrepancies are smaller than the unit-baseline discrepancies. The median absolute percentage discrepancy changes from 38.53% to 2.09% for Pantheon+, and from 36.39% to 2.96% for DES-Dovekie. Five of eight and three of seven predictions, respectively, lie within their marginal one-sigma intervals.

Independent reconstruction	Groups	Maximum |new − archived|
Pantheon+	8	7.06335×10^{-12}
DES-Dovekie	7	3.71803×10^{-12}

The new result is algebraic and numerical reconciliation: the explicit source-radius history gives the existing brightness calculation. The table reuses earlier observations and is not an additional independent empirical sample. Catalogue correlations and shared calibration information are not removed by reproducing the predictions.

13. The brightness trend in the same calculation

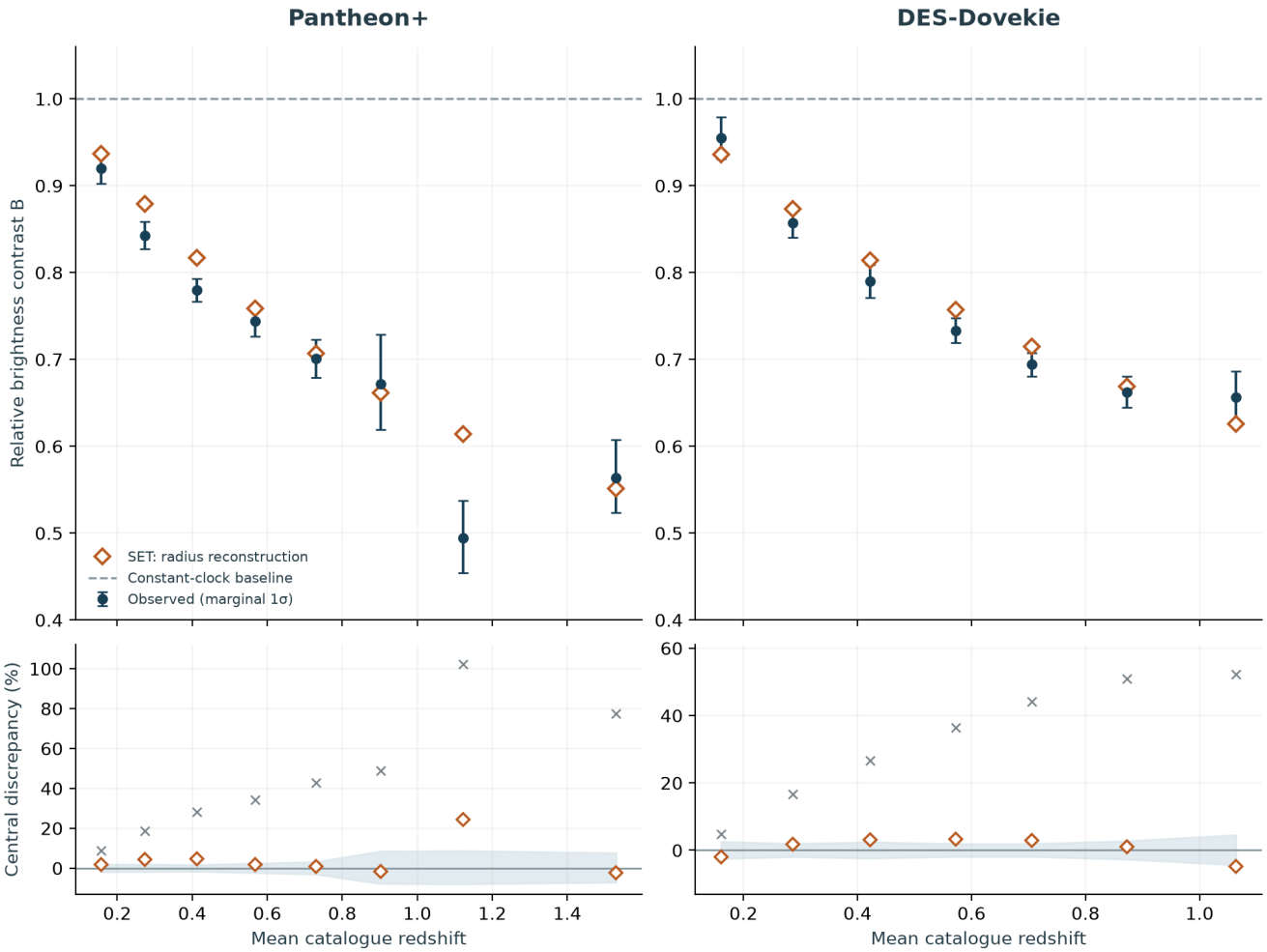


Figure 2. Archived grouped brightness contrasts and their reproduced SET predictions. Observations carry their published-analysis marginal one-sigma intervals. SET values are evaluated object by object before grouping. The horizontal line is the unit constant-clock baseline. Below, diamonds show SET discrepancies and gray crosses show baseline discrepancies. Shading shows the fractional observational one-sigma intervals about zero. This is a reproduction of the existing test, not a new fit or new sample.

The change is systematic across the two catalogue constructions: the source-clock history lowers the predicted relative brightness in the direction required by every central contrast. The table preserves the remaining differences, including the larger Pantheon+ residual near mean redshift 1.12.

The point of this derivation is that the brightness rule is now traceable through an explicit source radius, density, mass and clock history. The fifteen numerical predictions survive that change of mathematical route.

14. The initial source state and the front’s history

The exact integral also specifies the source-front history implied by the frozen reception state, when the zero-front branch is retained:

$$\frac{a_i}{a_0} = 0.2684965101, \qquad \frac{T_{Qi}}{T_{Q0}} = 0.1391258939 \tag{49}$$

$$T_{Qi} = 6.05953471 \text{ Gyr}, \qquad t_0 - t_i = 25.49724618 \text{ Gyr} \tag{50}$$

The first duration uses source-reference units; the second integrates the material clock dt = α dT_Q. They are model deductions from the chosen benchmark and history, not measured cosmic ages. The zero-radius front begins at finite representative-domain density and nonzero material scale factor.

Why the source start cannot be removed by a clock label

Define f(X) = −X/3 + √(1−2X²/3). On the positive-capacity interval,

$$f'(X) = -\frac{1}{3} - \frac{2X/3}{\sqrt{1 - 2X^2/3}} < 0 \tag{51}$$

The function crosses zero only at X_* = 3/√7. Below that point, a trajectory reaches X = 0 in finite backward ln T_Q. Above it, a trajectory reaches the capacity endpoint in finite backward ln T_Q. Hence only the fixed trajectory extends to every T_Q > 0 while keeping a nonnegative radius and a positive real clock.

$$u = \frac{6}{7}: \qquad t = \frac{T_Q}{\sqrt{7}}, \qquad R_h = 3ct, \qquad a \propto t^{2/3} \tag{52}$$

That self-similar trajectory has a constant clock factor and the constant-clock baseline distance shape. The varying branch used in the brightness calculation is a transient source-front history. Extending it to a formal zero-volume origin would require a different initial field state or a different physical regime; it is not achieved by shifting the clock’s origin.

Exact moving-volume bookkeeping

Integration from the retained launch to reception gives 54.7297% of V_h0 from the Q_normalized,h dt term and 45.2703% from cA_h dt. They sum to the full front volume. These are the two contributions in the moving-boundary identity, rather than two kinds of matter.

15. The connection to complete source energy

The source-radius derivation makes the next field question precise: can the common complete-energy prescription generate the retained clock-volume correspondence? It is useful to show the required relation explicitly, rather than leave that question as a general concern.

Within the previously examined flat homogeneous kinetic class, use an arbitrary coordinate λ and lapse N, with dt = N dλ. For the conserved material domain let E_b = M_Dc² and κ = 6πG/c². The reduced Lagrangian is

$$L = -\frac{V_\lambda^2}{4\kappa N V} - NE_b, \qquad V_\lambda \equiv \frac{dV}{d\lambda} \tag{53}$$

Vary N before fixing the clock. The constraint and its expanding branch are

$$\frac{V_\lambda^2}{4\kappa N^2 V} = E_b, \qquad \frac{dV}{dt} = \sqrt{4\kappa V E_b} = Q_b \tag{54}$$

Choosing a different coordinate lapse leaves this physical volume rate unchanged. The luminosity realization instead requires dV/dt = Q_b/α. Keeping that kinetic constraint, let E_complete denote the complete source/interaction contribution on its non-kinetic side. It excludes the kinetic term already displayed. The required rate then gives

$$E_{\text{complete}} = \frac{1}{4\kappa V}\left(\frac{Q_b}{\alpha}\right)^2 = \frac{E_b}{\alpha^2} \tag{55}$$

$$(1-u)E_{\text{complete}} = E_b \tag{56}$$

At the reception benchmark, E_complete/E_b = 4.16479720. This is the energy relation an actual source, field and stress configuration must supply within this class. The present derivation identifies the target; it does not construct that complete interacting state or add an unseen fluid to obtain it.

This distinction is physical. In a pure relabelling of the bare material solution, the lapse cancels between the scale-factor evolution and the null-distance integral. In the retained luminosity history it survives through H = H_b/α. The brightness response therefore asks for a source evolution, not merely a new name for time.

16. What this derivation establishes

The original intuition can now be stated in a calculable form. Matter inside a specified domain supplies Q according to one source law. A uniform central reading combines that domain's shells into S_c² = 3GM_h/R_h. Source growth and material dilution change that burden. The capacity relation assigns the corresponding clock progression, and the same clock enters the photon calculation.

Following those steps yields the radius equation, its exact burden history and the existing luminosity predictions. The calculation uses ordinary baryonic source inputs and introduces no fitted dark-energy density. It retains a physical correspondence between the source-volume clock and the capacity clock, together with the homogeneous source and front description stated in the derivation.

$$Q^2 = 24\pi GMV \quad \longrightarrow \quad T_Q,\ R_h(T_Q),\ u(T_Q) \quad \longrightarrow \quad D_L,\ B_{\mathrm{SET}} \tag{57}$$

The main advance is the explicit connection between those pieces. A successful brightness rule can now be examined as a definite source history with its mass, radius, density, clock and energy requirements visible. Within this realization the accumulating response is moderated by density evolution, and its initial causal source state matters. Those are physical deductions worth carrying forward.

Reproducibility and scope

The source-radius integration, exact-primitive derivative check, mass reconstruction and independent null integral were performed separately from the archived y(z) brightness code. The reception inputs, catalogue redshifts, reference groups and contrast weights were held fixed. The numerical comparison establishes equivalence of the two routes to the same predictions. It does not independently select the retained clock identification.

Sources and companion material

Echeverria, E. Space Emanation Theory - Integrated Rough Draft 4. The source and capacity framework, retained homogeneous luminosity prescription, catalogue protocol and observational references used here.

Echeverria, E. Space Emanation Theory - Source, Energy, Effective Area and Tidal Response, Rough Draft 1. Companion source-domain and energy-accounting analysis.

Research archive, 23 October 2026: "From the density source law to the luminosity history." Checked scripts: check_source_history.py and the separate source-radius luminosity reconstruction. Catalogue benchmark: verified-results.json from Space Emanation Theory - Dark Energy, Rough Draft 3. All fifteen archived contrasts are reproduced in this paper.